

# EchoChain – Power BI KPI Definitions

## Detailed specification for Task 4: Power BI dashboard planning

KPIs convert processed secondary-market, product lifecycle, component, and circularity data into measurable values that can be displayed and analyzed in Power BI.

### 1. Total Listings

The total number of secondary-market product listings collected through the web-scraping process. It indicates the volume of market data available for analysis. Example: 5,000 valid listings means Total Listings = 5,000.

### 2. Average Resale Price

The average secondary-market selling price of products in the dataset. Example: prices of ■20,000, ■25,000 and ■30,000 give an average of ■25,000. This KPI helps compare typical market value across products or SKUs.

### 3. Average Circularity Score

The average circularity score across analyzed products. It is intended to summarize potential for continued lifecycle activities such as reuse, repair, refurbishment or resale. The project's approved formula and weights should be used.

### 4. Component Failure Rate

The percentage of observed product/component records associated with a failure for a particular component, based on available project data. Example: 18 failures in 100 records gives 18%.

### 5. Depreciation

The reduction in product value when original/new price is compared with secondary-market resale price. Example: ■60,000 original and ■25,000 resale gives ■35,000 value reduction, or 58.33% depreciation.

### 6. Refurbishment Potential

An indicator for identifying products that may be suitable candidates for repair, refurbishment, reuse or resale using the project's approved market, lifecycle, component and cost-related rules.

## KPI Summary

| KPI                       | Purpose                                      |
|---------------------------|----------------------------------------------|
| Total Listings            | Shows secondary-market data volume           |
| Average Resale Price      | Shows typical secondary-market value         |
| Average Circularity Score | Shows overall circularity performance        |
| Component Failure Rate    | Highlights component failures                |
| Depreciation              | Shows product value reduction                |
| Refurbishment Potential   | Highlights possible refurbishment candidates |

## Recommended placement

Overview: Total Listings, Average Resale Price, Average Circularity Score, Average Failure Rate. Secondary Market: Total Listings, Average/Minimum/Maximum Price. Circularity & Components: Circularity Score, Failure Rate, Refurbishment Potential.

Data flow: Data Sources → Web Scraping → Databricks/Delta Lake → PySpark Cleaning & Transformation → Product/SKU Matching → Internal Data Integration → KPI Calculations → Power BI.